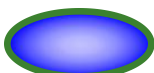
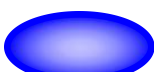
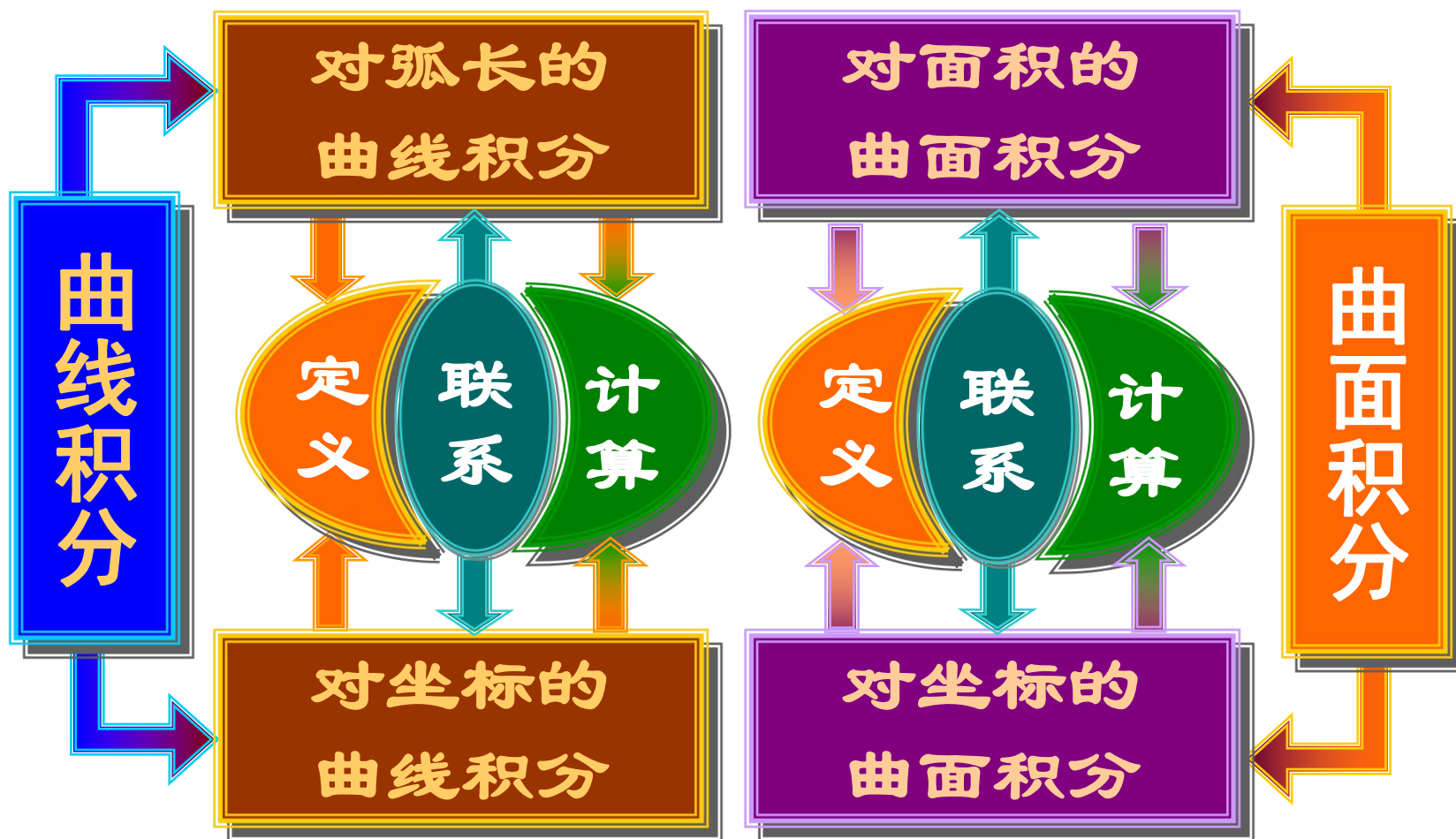


第 10 章 曲线积分与曲面积分

一、主要内容

-  (一) 曲线积分与曲面积分
-  (二) 各种积分之间的联系
-  (三) 场论初步

(一) 曲线积分与曲面积分



主要内容

一、概念

	曲线积分	
	对弧长的曲线积分	对坐标的曲线积分
定义	$\int_L f(x, y) ds = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta s_i$	$\begin{aligned} & \int_L P(x, y) dx + Q(x, y) dy \\ &= \lim_{\lambda \rightarrow 0} \sum_{i=1}^n [P(\xi_i, \eta_i) \Delta x_i + Q(\xi_i, \eta_i) \Delta y_i] \end{aligned}$

二、两类曲线积分之间的联系

联系	平面的情形	$\int_L P dx + Q dy = \int_L (P \cos \alpha + Q \cos \beta) ds$
	空间的情形	$\int_{\Gamma} P dx + Q dy + R dz = \int_{\Gamma} (P \cos \alpha + Q \cos \beta + R \cos \gamma) ds$

三、计算

■ 化为定积分

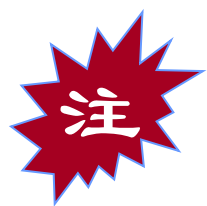
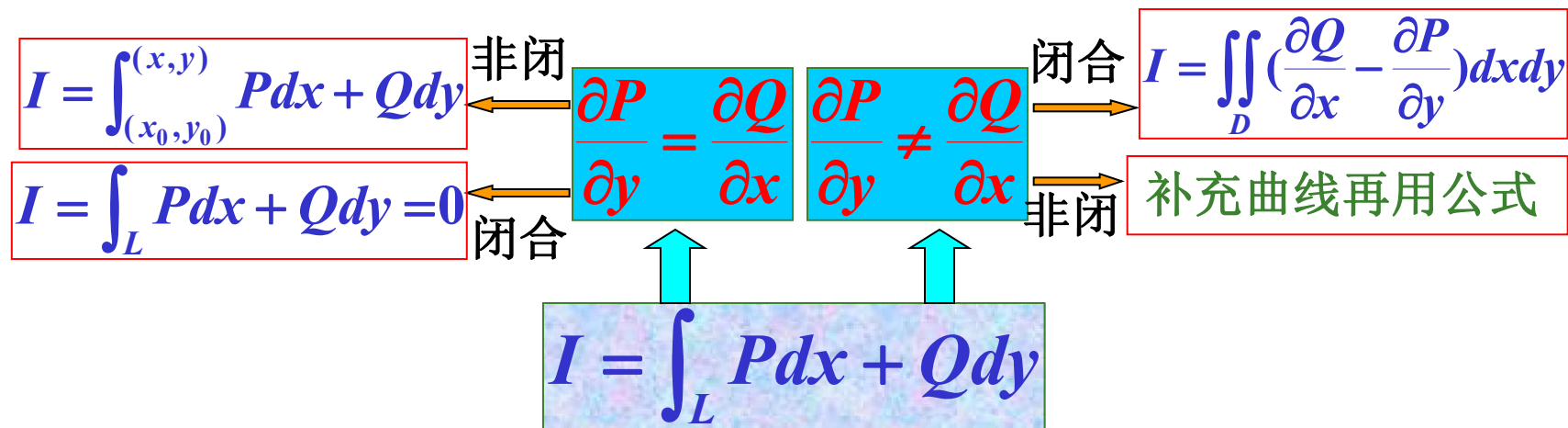
平面	$\int_L f(x, y) ds$ $= \int_{\alpha}^{\beta} f[\varphi, \psi] \sqrt{\varphi'^2 + \psi'^2} dt$	$\int_L P dx + Q dy$ $= \int_{\alpha}^{\beta} [P(\varphi, \psi) \varphi' + Q(\varphi, \psi) \psi'] dt$
空间	$\int_{\Gamma} f(x, y, z) ds$ $= \int_{\alpha}^{\beta} f[\varphi, \psi, \omega] \sqrt{\varphi'^2 + \psi'^2 + \omega'^2} dt$	$\int_L P dx + Q dy + R dz$ $= \int_{\alpha}^{\beta} [P(\varphi, \psi, \omega) \varphi' + Q(\varphi, \psi, \omega) \psi' + R(\varphi, \psi, \omega) \omega'] dt$
定限	$\alpha < \beta$	从始点→终点

四、Green公式的应用

与路径无关的四个等价命题

条件	在单连通开区域 D 上 $P(x, y), Q(x, y)$ 具有连续的一阶偏导数, 则以下四个命题成立.
等价命题	(1) 在 D 内 $\int_L Pdx + Qdy$ 与路径无关 (2) $\oint_C Pdx + Qdy = 0$, 闭曲线 $C \subset D$ (3) 在 D 内存在 $U(x, y)$ 使 $du = Pdx + Qdy$ (4) 在 D 内, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

■ 应用格林 (Green) 公式



若 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, 但被积函数在点 (x_0, y_0) 处不连续,

此时区域内有“洞”

此时, 绕包含这个“洞”在内的闭曲线其线积分不一定为 0, 但绕包含这个“洞”在内的闭曲线其线积分均相等。

五、线积分的应用

几何应用

• 计算曲线 L 的弧长 $\int_L 1 \cdot ds = s$

• 计算曲线所围区域的面积 $S = \frac{1}{2} \oint_L -ydx + xdy$

物理应用

• 平面金属杆 L 的质量 $M = \int_L \rho(x, y) ds$

• 空间金属杆 Γ 的质量 $M = \int_{\Gamma} \rho(x, y, z) ds$

•平面金属杆 L 的重心

$$\bar{x} = \frac{1}{M} \int_L x \rho(x, y) ds \quad \bar{y} = \frac{1}{M} \int_L y \rho(x, y) ds$$

•空间金属杆 Γ 的重心

$$\bar{x} = \frac{\int_{\Gamma} x \rho(x, y, z) ds}{M}, \quad \bar{y} = \frac{\int_{\Gamma} y \rho(x, y, z) ds}{M},$$

$$\bar{z} = \frac{\int_{\Gamma} z \rho(x, y, z) ds}{M}$$

•平面金属杆 L 绕坐标轴的转动惯量

$$I_x = \int_L y^2 \rho(x, y) ds \quad I_y = \int_L x^2 \rho(x, y) ds$$

•空间金属杆绕坐标轴的转动惯量

$$I_x = \int_{\Gamma} (x^2 + y^2) \rho(x, y, z) ds$$

$$I_y = \int_{\Gamma} (x^2 + z^2) \rho(x, y, z) ds$$

$$I_z = \int_{\Gamma} (x^2 + y^2) \rho(x, y, z) ds$$

•引力问题

•变力作功

一质点受变力作用沿曲线从 $A \rightarrow B$ 所作的功

$$W = \int_{\widehat{AB}} \vec{F} \cdot \vec{dr} = \int_{\widehat{AB}} p(x, y)dx + Q(x, y)dy$$

一、概念

	曲面积分	
	对面积的曲面积分	对坐标的曲面积分
定义	$\iint_{\Sigma} f(x, y, z) ds = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta s_i$	$\iint_{\Sigma} R(x, y, z) dx dy = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n R(\xi_i, \eta_i, \zeta_i) (\Delta S_i)_{xy}$

二、两类曲面积分之间的联系

联系	$\iint_{\Sigma} P dy dz + Q dz dx + R dx dy = \iint_{\Sigma} (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS$
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三、计算

■ 化为二重积分

$$\Sigma : z = z(x, y), (x, y) \in D_{xy}$$

$$\begin{aligned} \iint_{\Sigma} f(x, y, z) ds \\ = \iint_{D_{xy}} f[x, y, z(x, y)] \sqrt{1 + z_x^2 + z_y^2} dx dy \end{aligned}$$

(与侧无关)

$$\begin{aligned} \iint_{\Sigma} R(x, y, z) dx dy \\ = \pm \iint_{D_{xy}} R[x, y, z(x, y)] dx dy \end{aligned}$$

$\cos \gamma > 0$
 $\cos \gamma < 0$ (与侧有关)

■ 对坐标的曲面积分化为对面积的曲面积分

$$\iint_{\Sigma} P dy dz + Q dz dx + R dx dy = \iint_{\Sigma} (P \cos \alpha + Q \cos \beta + R \cos \gamma) dS$$

■ 应用高斯 (Guass) 公式

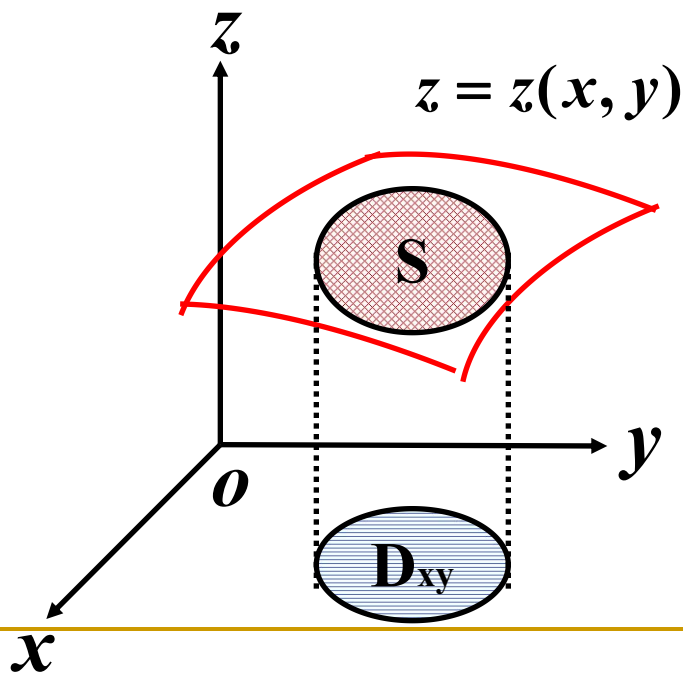
$$\oiint_{\Sigma} (\vec{A} \cdot \vec{n}) ds = \iiint_{\Omega} \operatorname{div} \vec{A} dv$$

$$\oiint_{\Sigma} P dydz + Q dzdx + R dxdy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dv$$

四、面积分的应用

■ 曲面面积的计算法

$$\begin{aligned} S &= \iint_{\Sigma} dS \\ &= \iint_{D_{xy}} \sqrt{1 + z_x^2 + z_y^2} dx dy \end{aligned}$$



■ 流量(通量)的计算法

设有向量场

$$\vec{A}(x, y, z) = P(x, y, z)\vec{i} + Q(x, y, z)\vec{j} + R(x, y, z)\vec{k}$$

向量场 $\vec{A}(x, y, z)$ 穿过曲面 Σ 指定侧的**通量**:

$$\Phi = \iint_{\Sigma} \vec{A} \cdot d\vec{S} = \iint_{\Sigma} \vec{A} \cdot \vec{n} dS = \iint_{\Sigma} P dydz + Q dzdx + R dxdy$$

五、场论初步

梯度 $\text{gradu} = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$

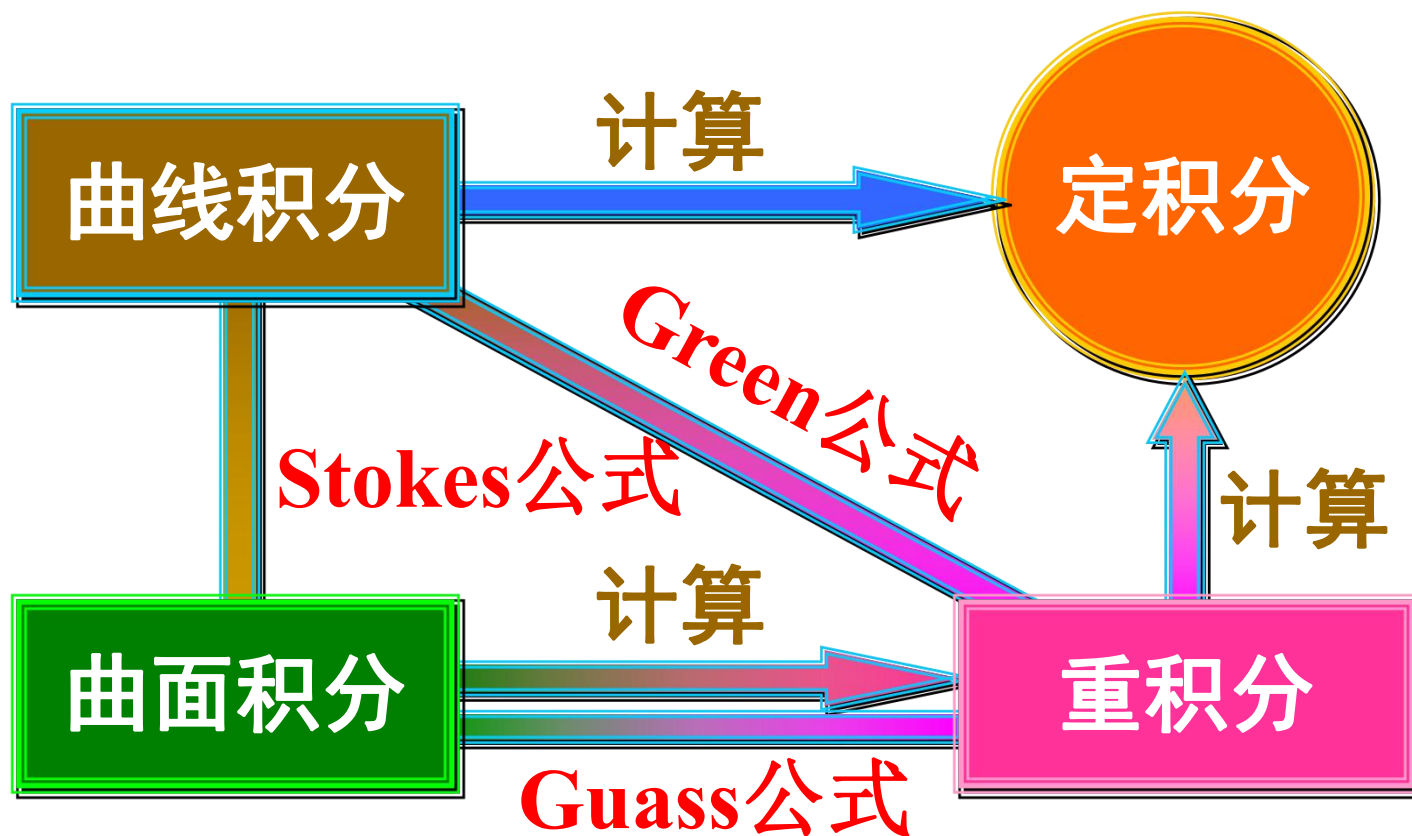
通量 $\Phi = \iint_{\Sigma} Pdydz + Qdzdx + Rdx dy$

散度 $\text{div} \vec{A} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$

环流量 $r = \oint_{\Gamma} Pdx + Qdy + Rdz$

旋度 $\text{rot} \vec{A} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}$

(二) 各种积分之间的联系



理论上的联系



1. 定积分与不定积分的联系

$$\int_a^b f(x)dx = F(b) - F(a) \quad (F'(x) = f(x))$$

牛顿--莱布尼茨公式

2. 二重积分与曲线积分的联系

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_L P dx + Q dy \quad (\text{沿 } L \text{ 的正向})$$

格林公式

3.三重积分与曲面积分的联系

$$\iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dv = \oiint_{\Sigma} P dydz + Q dzdx + R dxdy$$

高斯公式

4.曲面积分与曲线积分的联系

$$\begin{aligned} & \oiint_{\Sigma} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dydz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzdx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy \\ &= \oint_{\Gamma} P dx + Q dy + R dz \end{aligned}$$

斯托克斯公式

★ Green公式, Guass公式, Stokes公式之间的关系

$$\oint_L Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dxdy \quad \text{或} \quad \oint_L -Qdx + Pdy = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dxdy$$

$\vec{A}(M)$ 为平面向量场

$$\oint_L \vec{A} \cdot d\vec{s} = \iint_D (\text{rot} \vec{A} \cdot \vec{k}) dxdy$$

$$\oint_L (\vec{A} \cdot \vec{n}) ds = \iint_D \text{div} \vec{A} dxdy$$

推广

$\vec{A}(M)$ 为空间向量场

推广

$$\oint_{\Gamma} \vec{A} \cdot d\vec{S} = \iint_{\Sigma} (\text{rot} \vec{A} \cdot \vec{n}) dS$$

$$\oiint_{\Sigma} (\vec{A} \cdot \vec{n}) ds = \iiint_{\Omega} \text{div} \vec{A} dv$$

$$\oint_{\Gamma} Pdx + Qdy + Rdz$$

$$= \iint_{\Sigma} \begin{vmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$\oiint_{\Sigma} Pdydz + Qdzdx + Rdxdy$$

$$= \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dv$$

(三) 场论初步

梯度 $\text{gradu} = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k}$

通量 $\Phi = \iint_{\Sigma} Pdydz + Qdzdx + Rdx dy$

散度 $\text{div} \vec{A} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$

环流量 $\Gamma = \oint_{\Gamma} Pdx + Qdy + Rdz$

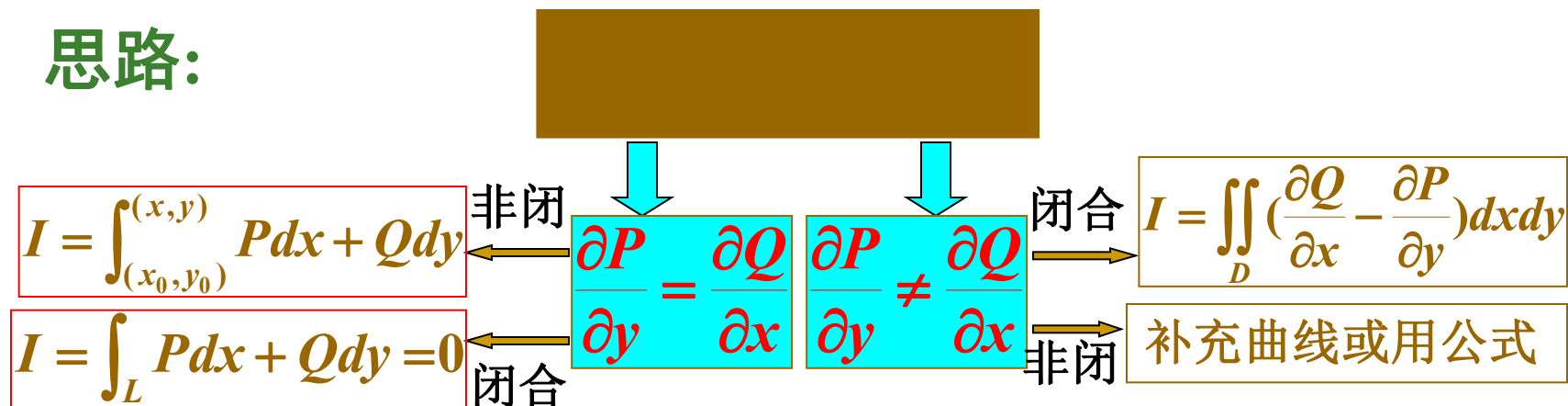
旋度 $\text{rot} \vec{A} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}$

二、典型例题

例 1 计算 $I = \int_L (x^2 + 2xy)dx + (x^2 + y^4)dy$,

其中 L 为由点 $O(0,0)$ 到点 $A(1,1)$ 的曲线 $y = \sin \frac{\pi}{2} x$.

思路:



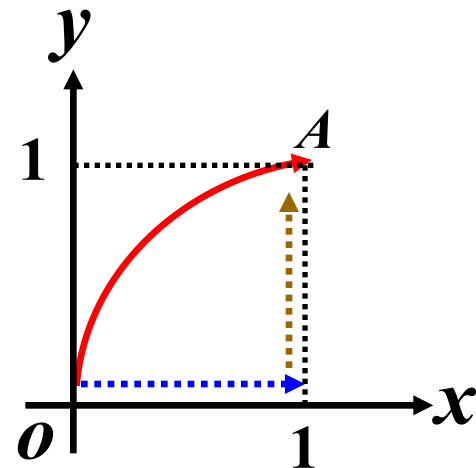
解 由 $I = \int (x^2 + 2xy)dx + (x^2 + y^4)dy$

知 $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2 + 2xy) = 2x$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^4) = 2x$$

即 $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x},$

$$\text{故原式} = \int_0^1 x^2 dx + \int_0^1 (1 + y^4) dy = \frac{23}{15}.$$



例 2 计算

$$I = \int_L (e^x \sin y - my) dx + (e^x \cos y - m) dy,$$

其中 L 为由点 $(a, 0)$ 到点 $(0, 0)$ 的上半圆周

$$x^2 + y^2 = ax, y \geq 0.$$

解 $\because \frac{\partial P}{\partial y} = \frac{\partial}{\partial y} (e^x \sin y - my) = e^x \cos y - m$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x} (e^x \cos y - m) = e^x \cos y$$

即 $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ (如下图)

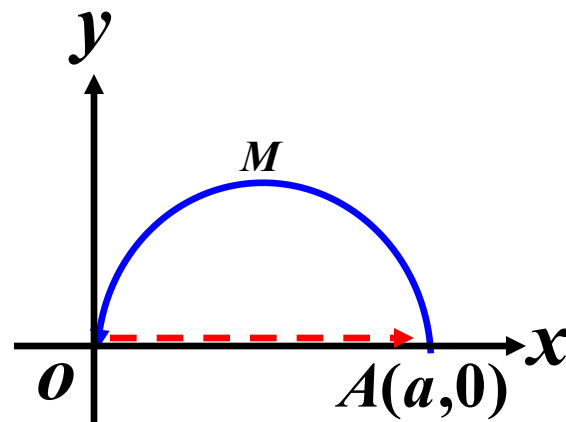
$$I = \int_{L+\overline{OA}} - \int_{\overline{OA}} = \oint_{AMOA} - \int_{\overline{OA}}$$

$$\oint_{AMOA} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

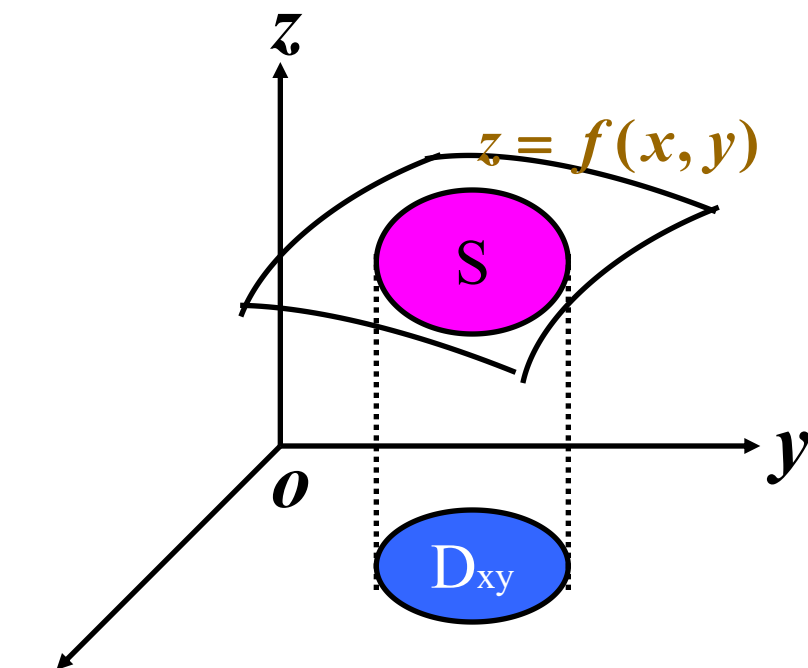
$$= m \iint_D dx dy = \frac{m}{8} \pi a^2,$$

$$\int_{\overline{OA}} = \int_0^a 0 \cdot dx + (e^x - m) \cdot 0 = 0,$$

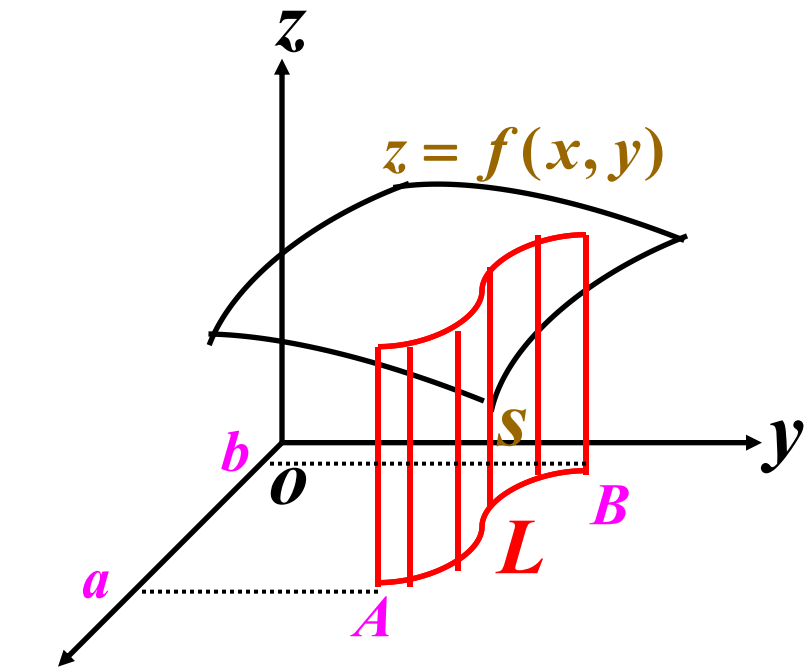
$$\therefore I = \oint_{AMOA} - \int_{\overline{OA}} = \frac{m}{8} \pi a^2 - 0 = \frac{m}{8} \pi a^2.$$



曲面面积的计算法



$$\begin{aligned}
 S &= \iint_{\Sigma} dS \\
 &= \iint_{D_{xy}} \sqrt{1 + z_x^2 + z_y^2} dx dy
 \end{aligned}$$

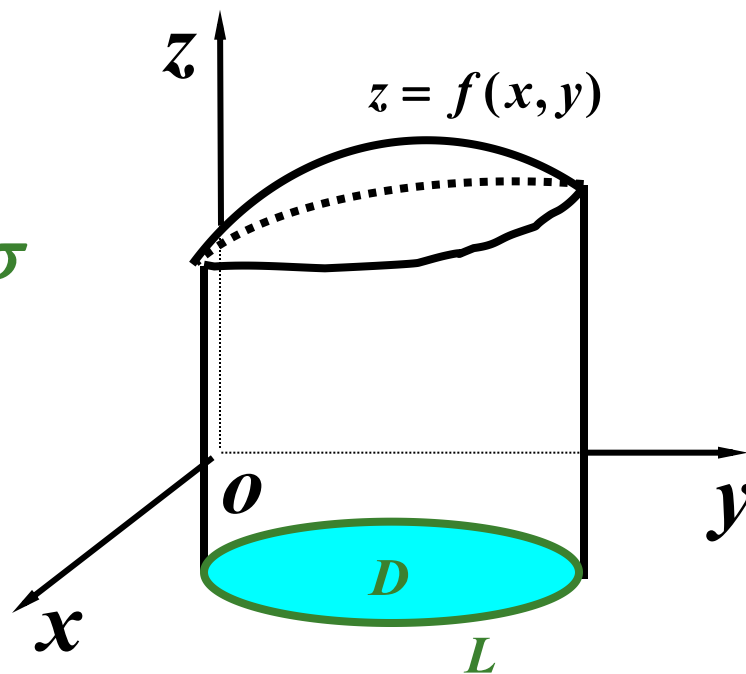


$$\begin{aligned}
 S &= \int_{L(A,B)} f(x, y) ds \\
 &= \int_a^b f(x, y) \sqrt{1 + y'^2} dx
 \end{aligned}$$

曲顶柱体的表面积

如图曲顶柱体,

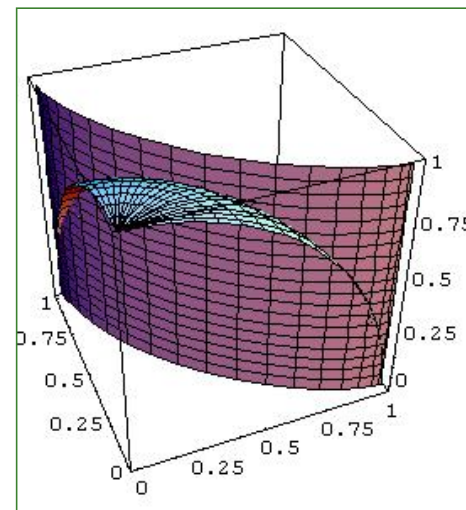
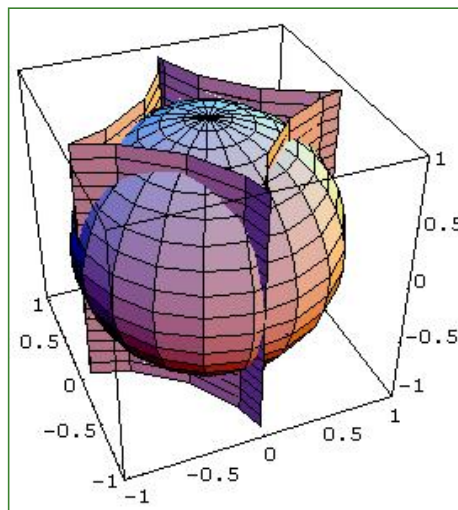
$$S = \iint_D (1 + \sqrt{1 + f_x^2 + f_y^2}) d\sigma \\ + \oint_L f(x, y) ds$$



例 3 求柱面 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$ 在球面 $x^2 + y^2 + z^2 = 1$ 内的侧面积.

解 由对称性

$$\begin{aligned} S &= 8 \int_L z ds \\ &= \int_L \sqrt{1 - x^2 - y^2} ds \end{aligned}$$



$$\therefore L: x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1, \text{ 参数方程为 } \begin{cases} x = \cos^3 t, \\ y = \sin^3 t, \end{cases} \quad (0 \leq t \leq \frac{\pi}{2})$$

$$ds = \sqrt{(x'_t)^2 + (y'_t)^2} dt = 3 \sin t \cos t dt,$$

$$S = 8 \int_0^{\frac{\pi}{2}} \sqrt{1 - \cos^6 t - \sin^6 t} 3 \sin t \cos t dt$$

$$= 24 \int_0^{\frac{\pi}{2}} \sqrt{3 \sin^2 t \cos^2 t} \sin t \cos t dt$$

$$= 24\sqrt{3} \int_0^{\frac{\pi}{2}} \sin^2 t \cos^2 t dt = \frac{3\sqrt{3}}{2} \pi.$$

例4 计算

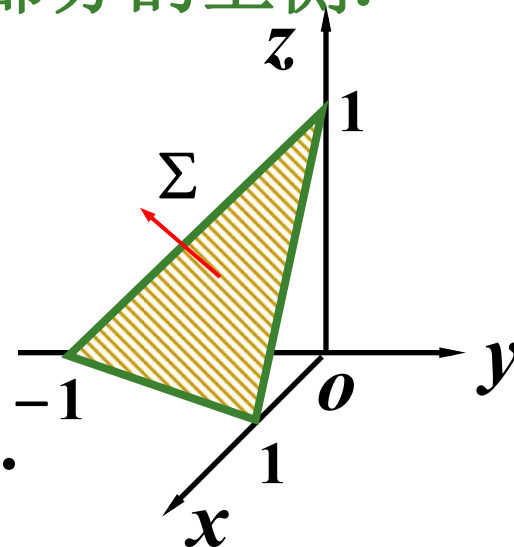
$$I = \iint_{\Sigma} [f(x, y, z) + x] dydz + [2f(x, y, z) + y] dzdx$$

$+ [f(x, y, z) + z] dxdy$, 其中 $f(x, y, z)$ 为连续函数,
 Σ 为平面 $x - y + z = 1$ 在第四卦限部分的上侧.

解 利用两类曲面积分之间的关系

$\therefore \Sigma$ 的法向量为 $\vec{n} = \{1, -1, 1\}$,

$$\therefore \cos \alpha = \frac{1}{\sqrt{3}}, \cos \beta = \frac{-1}{\sqrt{3}}, \cos \gamma = \frac{1}{\sqrt{3}}.$$



$$\begin{aligned}
 I &= \iint_{\Sigma} \left\{ \frac{1}{\sqrt{3}} [f(x, y, z) + x] \right. \\
 &\quad \left. - \frac{1}{\sqrt{3}} [2f(x, y, z) + y] + \frac{1}{\sqrt{3}} [f(x, y, z) + z] \right\} dS \\
 &= \frac{1}{\sqrt{3}} \iint_{\Sigma} (x - y + z) dS \\
 &= \frac{1}{\sqrt{3}} \iint_{D_{xy}} 1 \cdot \sqrt{3} dx dy = \frac{1}{2}.
 \end{aligned}$$

向量点积法

设 $\Sigma: z = f(x, y)$, 法向量为 $\{-f'_x, -f'_y, 1\}$,

$$I = \iint_{\Sigma} Pdydz + Qdzdx + Rdx dy$$

$$= \iint_{\Sigma} \{P, Q, R\} \cdot \{dydz, dzdx, dxdy\} = \iint_{\Sigma} \vec{A} \cdot \vec{n}^0 dS$$

$$= \iint_{\Sigma} \{P, Q, R\} \cdot \{-f'_x, -f'_y, 1\} dxdy$$

将 Σ 在 xoy 面投影 $\iint_{\Sigma} \{P, Q, R\} \cdot \{-f'_x, -f'_y, 1\} dxdy$.

例 5 计算曲面积分

$$I = \iint_{\Sigma} (8y+1)xdydz + 2(1-y^2)dzdx - 4yzdxdy,$$

其中 Σ 是由曲线 $\begin{cases} z = \sqrt{y-1} \\ x = 0 \end{cases} \quad (1 \leq y \leq 3)$ 绕 y 轴旋转一周

所成的曲面, 它的法向量与 y 轴正向的夹角恒大于 $\frac{\pi}{2}$.

解 $\begin{cases} z = \sqrt{y-1} \\ x = 0 \end{cases}$ 绕 y 轴旋转面方程为

$$y-1 = z^2 + x^2 \quad (\text{如下图})$$

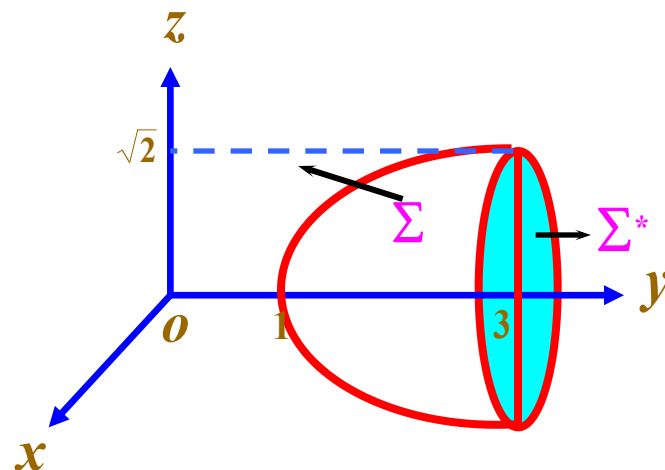
欲求 $I = \iint_{\Sigma} (8y + 1)xdydz + 2(1 - y^2)dzdx - 4yzdxdy$

且有 $I = \iint_{\Sigma + \Sigma^*} - \iint_{\Sigma^*}$

$$\iint_{\Sigma + \Sigma^*} = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dv$$

$$= \iiint_{\Omega} (8y + 1 - 4y - 4y) dv = \iiint_{\Omega} dv$$

$$= \iint_{D_{xz}} dx dz \int_{1+z^2+x^2}^3 dy = \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} \rho d\rho \int_{1+\rho^2}^3 dy$$



$$= 2\pi \int_0^{\sqrt{2}} (2\rho - \rho^3) d\rho = 2\pi,$$

$$\iint_{\Sigma^*} = 2 \iint_{\Sigma^*} (1 - 3^2) dz dx = -32\pi,$$

$$\text{故 } I = 2\pi - (-32\pi) = 34\pi.$$

例 6 求一个可微函数 $f(x, y)$ 满足 $f(0, 1) = 1$ ，并使曲线

$$\text{积分 } I_1 = \int_L (3xy^2 + x^3) dx + f(x, y) dy$$

和 $I_2 = \int_L f(x, y) dx + (3xy^2 + x^3) dy$ 都与积分路径无关.

解： 由积分与路径无关知： $\frac{\partial f}{\partial x} = 6xy$, $\frac{\partial f}{\partial y} = 3y^2 + 3x^2$

$\frac{\partial f}{\partial x} = 6xy$ 两边对 x 积分得 $f(x, y) = 3x^2y + C(y)$

$\frac{\partial f}{\partial y} = 3x^2 + C'(y) = 3y^2 + 3x^2$ 知 $C'(y) = 3y^2$,

即 $C(y) = y^3 + C$

由 $f(0, 1) = 1$ 知 $C = 0$, 故 $f(x, y) = 3x^2y + y^3$

例7 选取 λ , 使 $\int_L 2xy(x^4 + y^2)^\lambda dx - x^2(x^4 + y^2)^\lambda dy$
在右半平面 $x > 0$ 与路径无关,

并求 $\int_{(1,0)}^{(x,y)} 2xy(x^4 + y^2)^\lambda dx - x^2(x^4 + y^2)^\lambda dy$

解 令 $P = 2xy(x^4 + y^2)^\lambda, Q = -x^2(x^4 + y^2)^\lambda$

$$\because \int_L \text{与路径无关} \Leftrightarrow \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

$$\therefore 4x(x^4 + y^2)^\lambda(1 + \lambda) = 0, \text{解之得 } \lambda = -1$$

$$\int_{(1,0)}^{(x,y)} 2xy(x^4 + y^2)^\lambda dx - x^2(x^4 + y^2)^\lambda dy$$

$$= \int_{(1,0)}^{(x,y)} \frac{2xydx - x^2dy}{x^4 + y^2} = \int_1^x \frac{2x \cdot 0 dx}{x^4 + 0^2} + \int_0^y \frac{-x^2 dy}{x^4 + y^2} = -\arctan \frac{y}{x^2}$$

例 8 设在上半平面 $D = \{(x, y) \mid y > 0\}$ 内, 函数 $f(x, y)$ 具有连续偏导数, 且对任意的 $t > 0$ 都有 $f(tx, ty) = t^{-2} f(x, y)$. 证明: 对 D 内的任意分段光滑的有向简单闭曲线 L , 都有 $\oint_L yf(x, y)dx - xf(x, y)dy = 0$.

利用曲线积分与路径无关的条件

齐次函数求导性质

解: 令 $P(x, y) = yf(x, y), Q(x, y) = -xf(x, y)$, 则

$$\frac{\partial Q}{\partial x} = -f(x, y) - xf'_x(x, y), \frac{\partial P}{\partial y} = f(x, y) + yf'_y(x, y).$$

将 $f(tx, ty) = t^{-2} f(x, y)$ 两边对 t 求导得

$$xf'_x(tx, ty) + yf'_y(tx, ty) = -2t^{-3} f(x, y).$$

令 $t = 1$, 则 $xf'_x(x, y) + yf'_y(x, y) = -2f(x, y)$

则由此可得 $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$. $\therefore \oint_L yf(x, y)dx - xf(x, y)dy = 0$.

例 9 已知曲线积分 $\oint_C \frac{xdy - ydx}{2y^2 + \varphi(x)} \equiv A$ (A 为常数), 其中函数 $\varphi(x)$ 具有

连续导数, 且 $\varphi(1)=1$, C 是围绕原点一周的任一正向闭曲线,

(1) 证明在任一不包含原点的单连通区域内, 曲线积分

$\int_C \frac{xdy - ydx}{2y^2 + \varphi(x)}$ 与路径无关;

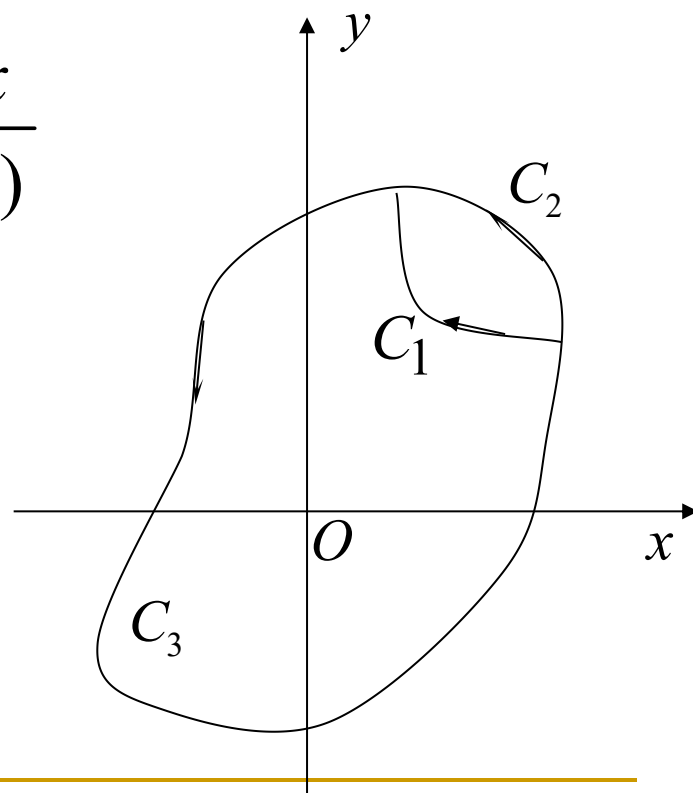
(2) 求函数 $\varphi(x)$ 的表达式, 并求 A 的值.

积分与路径无关的问题!

解：（1）在不包含原点的单连通区域内，任取两条具有相同起点与终点的曲线

C_1 和 C_2 ，再补上一条光滑曲线 C_3 ，使 $C_1 + C_3$ 和 $C_2 + C_3$ 成为围绕原点的正向闭曲线，由题意知

$$\oint_{C_1 + C_3} \frac{xdy - ydx}{2y^2 + \varphi(x)} = \oint_{C_2 + C_3} \frac{xdy - ydx}{2y^2 + \varphi(x)}$$
$$\Rightarrow \int_{C_1} \frac{xdy - ydx}{2y^2 + \varphi(x)} = \int_{C_2} \frac{xdy - ydx}{2y^2 + \varphi(x)}$$



即在任一不包含原点的单连通区域内，
曲线积分与路径无关。

$$(2) \quad P = \frac{-y}{2y^2 + \varphi(x)}, \quad Q = \frac{x}{2y^2 + \varphi(x)}$$

$$\frac{\partial P}{\partial y} = \frac{2y^2 - \varphi(x)}{[2y^2 + \varphi(x)]^2} \quad \frac{\partial Q}{\partial x} = \frac{2y^2 + \varphi(x) - x\varphi'(x)}{[2y^2 + \varphi(x)]^2}$$

由 (1) 知, 当 $(x, y) \neq (0, 0)$ 时,

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow 2y^2 - \varphi(x) = 2y^2 + \varphi(x) - x\varphi'(x)$$

$$\varphi'(x) - \frac{2}{x}\varphi(x) = 0 \Rightarrow \varphi(x) = Cx^2$$

由 $\varphi(1) = 1$ 得, $C = 1$, 即 $\varphi(x) = x^2$,

取 C 为正向椭圆 $x^2 + 2y^2 = 1$, 则

$$A = \oint_C \frac{xdy - ydx}{x^2 + 2y^2} = \oint_C xdy - ydx = 2 \iint_D dx dy = 2\pi \cdot 1 \cdot \frac{1}{\sqrt{2}} = \sqrt{2}\pi$$

例10 设 $u(x,y,z)$ 和 $v(x,y,z)$ 在闭区域上具有一阶及二阶连续偏导数, 证明:

$$\iiint_{\Omega} u \Delta v dx dy dz = \oiint_{\Sigma} u \frac{\partial v}{\partial n} dS - \iiint_{\Omega} \nabla u \cdot \nabla v dv$$

其中 Σ 是闭区域的整个边界曲面, $\frac{\partial v}{\partial n}$ 为函数 $v(x,y,z)$

沿 Σ 的外法线方向的方向导数,

符号 $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ 称为 *Laplace* 算子。

符号 $\nabla = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$ 称为 *Halmiton* 算子。

$$\text{gradu} \cdot \text{grad} v = \nabla u \cdot \nabla v = \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial v}{\partial z}$$

这个公式叫做格林第一公式

证明：设外法线单位向量 $\vec{n} = \{\cos \alpha, \cos \beta, \cos \gamma\}$

$$\begin{aligned}\iint_{\Sigma} u \frac{\partial v}{\partial n} dS &= \iint_{\Sigma} u(v_x' \cos \alpha + v_y' \cos \beta + v_z' \cos \gamma) dS \\&= \iiint_{\Omega} \left(\frac{\partial(uv_x')}{\partial x} + \frac{\partial(uv_y')}{\partial y} + \frac{\partial(uv_z')}{\partial z} \right) dv \\&= \iiint_{\Omega} u(v_{xx}'' + v_{yy}'' + v_{zz}'') + (u_x' v_x' + u_y' v_y' + u_z' v_z') dv \\&= \iiint_{\Omega} u \Delta v dx dy dz + \iiint_{\Omega} \nabla u \cdot \nabla v dv\end{aligned}$$

移项得

$$\iiint_{\Omega} u \Delta v dx dy dz = \iint_{\Sigma} u \frac{\partial v}{\partial n} dS - \iiint_{\Omega} \nabla u \cdot \nabla v dv$$

举

例

例1 求 $I = \int_{\Gamma} x^2 ds$,

其中 Γ 为圆周
$$\begin{cases} x^2 + y^2 + z^2 = a^2, \\ x + y + z = 0. \end{cases}$$

解法一 由对称性, 知 $\int_{\Gamma} x^2 ds = \int_{\Gamma} y^2 ds = \int_{\Gamma} z^2 ds$.

$$\begin{aligned} \text{故 } I &= \frac{1}{3} \int_{\Gamma} (x^2 + y^2 + z^2) ds \\ &= \frac{a^2}{3} \int_{\Gamma} ds = \frac{2\pi a^3}{3}. \quad (2\pi a = \int_{\Gamma} ds, \text{球面大圆周长}) \end{aligned}$$

解法二 将 $y = -(x + z)$ 代入 $x^2 + y^2 + z^2 = a^2$ 得

$$x^2 + (x + z)^2 + z^2 = a^2 \Rightarrow ds = \sqrt{x'^2 + y'^2 + z'^2} dt$$

$$\frac{3}{2}x^2 + 2(z + \frac{x}{2})^2 = a^2, \text{ 则} \quad = a dt$$

$$L : \begin{cases} \frac{3}{2}x^2 + 2(z + \frac{x}{2})^2 = a^2 \\ x + y + z = 0 \end{cases} \quad \int_L x^2 ds = \int_0^{2\pi} \frac{2}{3}a^2 \cos^2 t a dt$$
$$= \frac{2\pi}{3}a^3$$

化为参数方程

$$\begin{cases} x = \sqrt{\frac{2}{3}}a \cos t, \\ z + \frac{x}{2} = \frac{a}{\sqrt{2}} \sin t \Rightarrow z = \frac{a}{\sqrt{2}} (\sin t - \frac{1}{\sqrt{3}} \cos t), \\ y = -(x + z) = -\frac{a}{\sqrt{2}} (\sin t + \frac{1}{\sqrt{3}} \cos t) \end{cases}$$

例2 计算： $\int_{\widehat{OA}} e^x (2 - \cos y) dx + e^x (\sin y - y) dy$

$\widehat{OA}$ 为从点 $O(0,0)$ 经正弦曲线 $y = \sin x$ 到点 $A(\pi,0)$ 的弧段.

解法一

$$\begin{aligned}\int_{\widehat{OA}} &= \oint_{\widehat{OAO}} - \int_{\overline{AO}} = - \iint_D -e^x y dx dy + \int_{\overline{OA}} \\ &= \int_0^\pi dx \int_0^{\sin x} e^x y dy + \int_0^\pi e^x dx \\ &= \frac{1}{5} (e^\pi - 1) + (e^\pi - 1) = 1 \frac{1}{5} (e^\pi - 1)\end{aligned}$$

解法二 $\int_{\widehat{OA}} e^x (2 - \cos y) dx + e^x (\sin y - y) dy$

$$= \int_{\widehat{OA}} e^x (-\cos y) dx + e^x \sin y dy + \int_{\widehat{OA}} 2e^x dx - ye^x dy$$

$\because \int_{\widehat{OA}} e^x (-\cos y) dx + e^x \sin y dy$ 与路径无关

$\Downarrow$

$$= \int_0^\pi e^x (-\cos 0) dx + e^x \sin 0 dy$$

$$+ \int_0^\pi 2e^x dx - (\sin x)e^x d(\sin x)$$

$$= 1 \frac{1}{5} (e^\pi - 1)$$

例3 在变力 $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$ 的作用下, 质点由原点沿直线运动到椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 上第一卦限的点 $M(\xi, \eta, \zeta)$, 问 ξ, η, ζ 取何值时, 力所做的功 w 最大? 并求出的最大值 .

解 直线段 $\overline{OM} : x = \xi t, y = \eta t, z = \zeta t, t : 0 \rightarrow 1$

$$w = \int_{\overline{OM}} yzdx + zxdy + xydz = \int_0^1 3\xi\eta\zeta t^2 dt = \xi\eta\zeta$$

下面求 $w = \xi\eta\zeta$ 在条件 $\frac{\xi^2}{a^2} + \frac{\eta^2}{b^2} + \frac{\zeta^2}{c^2} = 1$

$(\xi \geq 0, \eta \geq 0, \zeta \geq 0)$ 下的最大值

令 $f(\xi, \eta, \zeta) = \xi\eta\zeta + \lambda(1 - \frac{\xi^2}{a^2} - \frac{\eta^2}{b^2} - \frac{\zeta^2}{c^2})$

由 $\frac{\partial f}{\partial \xi} = 0, \frac{\partial f}{\partial \eta} = 0, \frac{\partial f}{\partial \zeta} = 0$, 得

$$\eta\zeta = \frac{2\lambda}{a^2}\xi, \xi\zeta = \frac{2\lambda}{b^2}\eta, \xi\eta = \frac{2\lambda}{c^2}\zeta$$

从而 $\frac{\xi^2}{a^2} = \frac{\eta^2}{b^2} = \frac{\zeta^2}{c^2}$ 即得

$$\frac{\xi^2}{a^2} = \frac{\eta^2}{b^2} = \frac{\zeta^2}{c^2} = \frac{1}{3} \Rightarrow \xi = \frac{a}{\sqrt{3}}, \eta = \frac{b}{\sqrt{3}}, \zeta = \frac{c}{\sqrt{3}}$$

由问题的实际意义知 $w_{\max} = \frac{\sqrt{3}}{9}abc$

例4 $\oint_c (z - y)dx + (x - z)dy + (x - y)dz$

其中: $c: \begin{cases} x^2 + y^2 = 1 \\ x - y + z = 2 \end{cases}$ 从正 z 轴方向
往负 z 轴方向看是顺时针.

解法 1 将曲线 c 参数化

令 $x = \cos \theta, y = \sin \theta, z = 2 - x + y = 2 - \cos \theta + \sin \theta$
 $\theta: 2\pi \rightarrow 0$

$$\oint_c (z - y)dx + (x - z)dy + (x - y)dz =$$

$$= \int_{2\pi}^0 [-2(\cos \theta + \sin \theta) + 3 \cos^2 \theta - \sin^2 \theta] = -2\pi$$

解法2 用stokes公式

取 $S: x - y + z = 2$ 上以 L 为边界的有限部分。
其法向量与 z 轴的正向夹角为钝角。

$D: x^2 + y^2 \leq 1$ 为 S 在 xoy 面上的投影，则

$$\begin{aligned} & \oint_L (z - y)dx + (x - z)dy + (x - y)dz \\ &= \iint_S \begin{vmatrix} dydz & dzdx & dxdy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z - y & x - z & x - y \end{vmatrix} = \iint_S 2dxdy \\ &= \iint_D -2dxdy = -2\pi \end{aligned}$$

例5 计算:

$$I = \oint_L \frac{xdy - ydx}{4x^2 + y^2} \quad L: (x-1)^2 + y^2 = R^2 (R > 0) \text{取逆时针}。$$

解

$$P = \frac{-y}{4x^2 + y^2}, Q = \frac{x}{4x^2 + y^2},$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} = \frac{y^2 - 4x^2}{(4x^2 + y^2)^2} (x^2 + y^2 \neq 0)$$

$$(1) R < 1 \quad I = 0$$

(2) $R > 1$ 作充分小的椭圆: $L_\varepsilon: 4x^2 + y^2 = \varepsilon^2$ (取逆时针)。

$$I = \oint_L \frac{xdy - ydx}{4x^2 + y^2} = \oint_{L_\varepsilon} \frac{xdy - ydx}{4x^2 + y^2} = \oint_{L_\varepsilon} \frac{xdy - ydx}{\varepsilon^2} = \pi$$

课堂练习

一、填空题

1. 设 $L: y = -\sqrt{1-x^2}$, 则 $\int_L \frac{1}{x^2 + y^2} ds = \underline{\pi}$

2. 设 $L: x^2 + \frac{y^2}{4} = 1$, 其周长为 a ,

则 $\oint_L (2xy + y^3 + 4x^2 + y^2) ds = \underline{4a}$

3. $L: x^2 + (y-1)^2 = 4$ 正向闭路, 则 $\oint_L \frac{xdy - ydx}{x^2 + (y-1)^2}$
 $= \frac{1}{4} \oint_L xdy - ydx = \frac{1}{4} \iint_D 2dxdy = 2\pi$

4. 已知: $\frac{(x+ay)dx + ydy}{(x+y)^2}$ 为某函数的全微分, 则 $a = \underline{2}$

二、计算下列各题

1 $\oint_L |xy| ds$ L : 圆周 $x^2 + y^2 = a^2 (a > 0)$

解 原式 $= \int_0^{2\pi} |a^2 \sin t \cos t| a dt$
 $= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |a^2 \sin t \cos t| a dt$
 $= 4 \int_0^{\frac{\pi}{2}} a^2 \sin t \cos t a dt = 2a^3$

又解 由对称性, 设 $\frac{L}{4}$: $x^2 + y^2 = a^2, x \geq 0, y \geq 0$

$$\text{原式} = 4 \int_{\frac{L}{4}} xy ds = 4 \int_0^{\frac{\pi}{2}} a^2 \sin t \cos t a dt = 2a^3$$

又解 原式 = $\int_0^{2\pi} |a^2 \sin t \cos t| a dt = a^3 \int_0^{2\pi} |\sin t \cos t| dt$

$$= a^3 \left\{ \int_0^{\frac{\pi}{2}} \sin t \cos t dt + \int_{\frac{\pi}{2}}^{\pi} -\sin t \cos t dt \right.$$

$$\left. + \int_{\pi}^{\frac{3\pi}{2}} \sin t \cos t dt + \int_{\frac{3\pi}{2}}^{2\pi} -\sin t \cos t dt \right\}$$

$$= a^3 \left\{ \frac{1}{2} \sin^2 t \Big|_0^{\frac{\pi}{2}} + \left(-\frac{1}{2} \sin^2 t \right) \Big|_{\frac{\pi}{2}}^{\pi} + \frac{\sin^2 t}{2} \Big|_{\pi}^{\frac{3\pi}{2}} + \left(-\sin^2 t \right) \Big|_{\frac{3\pi}{2}}^{2\pi} \right\}$$

$$= a^3 \left\{ \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right\} = 2a^3$$

$$2 \int_L \sqrt{x^2 + y^2} ds \quad L: x^2 + y^2 = ax$$

解法一 令 $x - \frac{a}{2} = \frac{a}{2} \cos t, \quad y = \frac{a}{2} \sin t$

$$x' = -\frac{a}{2} \sin t, \quad y' = \frac{a}{2} \cos t, \quad ds = \frac{a}{2} dt,$$

$$t: 0 \rightarrow 2\pi$$

$$\therefore \int_L \sqrt{x^2 + y^2} ds = \int_L \sqrt{ax} ds = \int_0^{2\pi} \sqrt{\frac{a}{2}(1 + \cos t)} \frac{a}{2} dt$$

$$= \int_0^{2\pi} a \left| \cos \frac{t}{2} \right| \frac{a}{2} dt = 2 \frac{a^2}{2} \left[2 \sin \frac{t}{2} \right]_0^\pi = 2a^2$$

解法二 $L : r = r(\theta) = a \cos \theta, r' = -a \sin \theta$

$$x = r(\theta) \cos \theta = a \cos^2 \theta$$

$$y = r(\theta) \sin \theta = a \cos \theta \sin \theta$$

$$\int_L \sqrt{x^2 + y^2} ds = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a \cos \theta \sqrt{r^2 + r'^2} d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a \cos \theta \sqrt{a} d\theta = 2a^2$$

解法三 $L: y = \pm \sqrt{ax - x^2}$

$$\int_L \sqrt{x^2 + y^2} ds$$

$$= 2 \int_0^a \sqrt{ax} \sqrt{1 + \left[\frac{a-2x}{2\sqrt{ax-x^2}} \right]^2} dx = \int_0^a \sqrt{ax} \frac{a}{\sqrt{ax-x^2}} dx$$

$$\left(\stackrel{\sqrt{x}=t}{=} \int_0^{\sqrt{a}} \frac{\sqrt{aat}}{\sqrt{at^2-t^4}} 2t dt = \int_0^{\sqrt{a}} \frac{2\sqrt{aat}}{\sqrt{a-t^2}} dt = \int_0^{\sqrt{a}} \frac{a\sqrt{a} dt^2}{\sqrt{a-t^2}} \right.$$

$$= [-2\sqrt{a-t^2}]_0^{\sqrt{a}} a\sqrt{a} = 2a^2)$$

$$= \int_0^a \frac{\sqrt{aa}}{\sqrt{a-x}} dx = \sqrt{aa} [-2\sqrt{a-x}]_0^a = 2a^2$$

3. 计算：
$$\int_L \left(\sin \frac{x}{y} + \frac{x}{y} \cos \frac{x}{y} \right) dx - \frac{x^2}{y^2} \cos \frac{x}{y} dy$$

L ：从点 $A(\pi, 1)$ 经过 $x = \pi(y-2)^2$ 到点 $B(\pi, 3)$ 的弧段

解 $\because \frac{\partial Q}{\partial x} = -\frac{2x}{y^2} \cos \frac{x}{y} + \frac{x^2}{y^3} \sin \frac{x}{y} = \frac{\partial P}{\partial x} (y > 0)$

$\therefore$ 曲线积分与路径无关，则

$$\begin{aligned} \int_L &= \int_{(\pi, 1)}^{(\pi, 3)} \left(\sin \frac{x}{y} + \frac{x}{y} \cos \frac{x}{y} \right) dx - \frac{x^2}{y^2} \cos \frac{x}{y} dy \\ &= \int_1^3 -\frac{\pi^2}{y^2} \cos \frac{\pi}{y} dy = \frac{\sqrt{3}}{2} \pi \end{aligned}$$

$$4. \int_C \frac{(x+y)dx - (x-y)dy}{x^2 + y^2} \quad \text{其中 } C :$$

(1) 正向圆周 $x^2 + y^2 = a^2$

(2) $a^2 \leq x^2 + y^2 \leq b^2$ ($b > a > 0$, 取正向)

(3) $A(-\pi, \pi), B(\pi, -\pi)$ $\widehat{AB} : y = \pi \cos x$

(4) 正方形 $|x| + |y| = 1$ 的正向

解(1)

法一 $x = a \cos t \quad y = a \sin t$

$$\text{原式} = \int_0^{2\pi} \frac{a(\cos t + \sin t)(-a \sin t) - a(\cos t - \sin t)(a \cos t)}{a^2} dt$$

$$= \int_0^{2\pi} -dt = -2\pi$$

法二

$$\begin{aligned}& \int_C \frac{(x+y)dx - (x-y)dy}{x^2 + y^2} \\&= \frac{1}{a^2} \int_C (x+y)dx - (x-y)dy \\&= \frac{1}{a^2} \iint_D (-1-1)d\sigma = -2\pi\end{aligned}$$

解(2)

$$\begin{aligned}\therefore \frac{\partial P}{\partial y} &= \frac{x^2 + y^2 - 2y(x+y)}{(x^2 + y^2)^2} \\&= \frac{x^2 - 2xy - y^2}{(x^2 + y^2)^2} = \frac{\partial Q}{\partial x} \\ \therefore \text{原式} &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = 0\end{aligned}$$

解(3) 添加辅助线如图

$$\because \oint_{\widehat{AB} + \overline{BA} + l} Pdx + Qdy$$

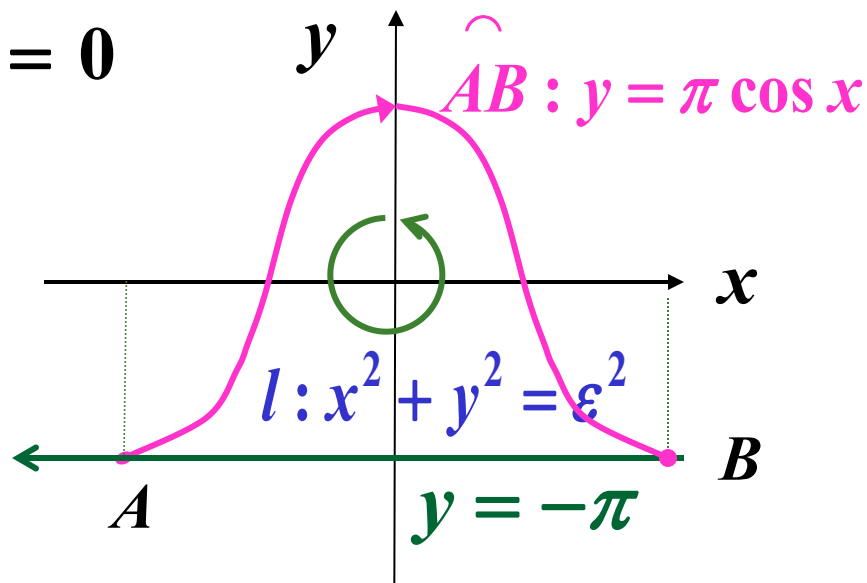
$$= - \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\sigma = 0$$

$$\therefore \int_{\widehat{AB}} = \int_{\overline{AB}} - \int_l$$

由(1)有 $\int_l = -2\pi$

$$\text{又} \int_{\overline{AB}} = \int_{-\pi}^{\pi} \frac{x - \pi}{x^2 + \pi^2} dx$$

$$= -2 \arctan \frac{x}{\pi} \Big|_0^{\pi} = -\frac{\pi}{2}$$



$$\therefore \int_{\widehat{AB}} = \int_{\overline{AB}} - \int_l = \frac{3\pi}{2}$$

解(4) 正方形 $C : |x| + |y| = 1$ 的正向

添加辅助线 $-l : l$ 同第(3)小题, 则

$$\oint_C = -\oint_{-l} = \oint_l - 2\pi$$

三、设 $\vec{F} = \{x + 2y + 4, 4x - 2y, 3x + z\}$,

试求单位质点 M 沿椭圆 C :

$$(3x + 2y - 5)^2 + (x - y + 1)^2 = a^2, \quad z = 4(a > 0)$$

移动一周 (取正向), 力 $\vec{F}$ 所作的功 W .

解 $W = \int_C \vec{F} \cdot \vec{\tau} ds = \oint_C (x + 2y + 4)dx + (4x - 2y)dy$

$$= \iint_D (4 - 2) dxdy = 2 \iint_D dxdy = 2 \iint_{D'} |J| dudv = \frac{2}{5} \iint_{D'} dudv$$

$$= \frac{2}{5} \pi a^2 \quad \text{其中: } |J| = \frac{1}{\frac{\partial(u,v)}{\partial(x,y)}} = \frac{1}{\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix}} = \frac{1}{-3-2} = \left| -\frac{1}{5} \right|$$

$$\begin{cases} u = 3x + 2y - 5 \\ v = x - y + 1 \end{cases} \quad D': u^2 + v^2 \leq a^2$$

四、过 $O(0,0)$ 和 $A(\pi,0)$ 的曲线族 $y = a \sin x (a > 0)$ 中求一条曲线 l 使该曲线从 O 到 A 积分

$$I = \int_L (1 + y^3) dx + (2x + y) dy \text{ 最小.}$$

解
$$\begin{aligned} I(a) &= \int_L (1 + y^3) dx + (2x + y) dy \\ &= \int_0^\pi [(1 + a^3 \sin^3 x) + (2x + a \sin x) a \cos x] dx \\ &= \pi - 4a + \frac{4}{3} a^3 \end{aligned}$$

$$I'(a) = 4a^2 - 4 \Rightarrow a = 1, a = -1(\text{舍去}),$$

$\therefore$ 当 $a = 1$ 时, 即曲线 $y = \sin x$ 使曲线积分 I 最小.

例 计算 $I = \iint_{\Sigma} ydydz - xdzdx + z^2dxdy$, 其中 Σ 为
锥面 $z = \sqrt{x^2 + y^2}$ 被平面 $z = 1, z = 2$ 所截部分的外侧.

解 利用两类曲面积分之间的关系

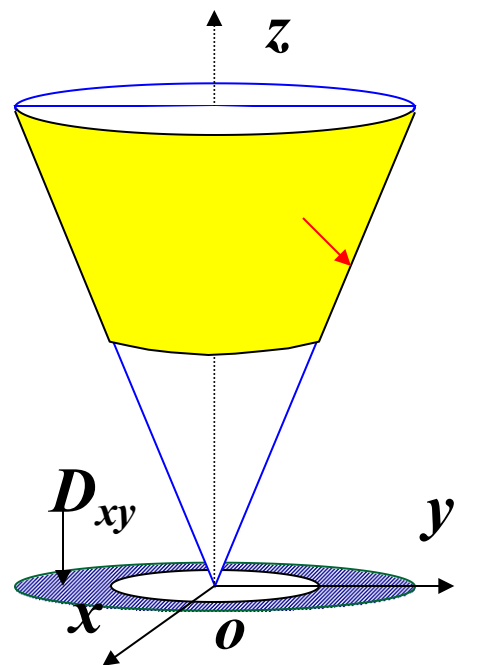
$$\because f'_x = \frac{x}{\sqrt{x^2 + y^2}}, f'_y = \frac{y}{\sqrt{x^2 + y^2}},$$

$$\sqrt{1 + (z'_x)^2 + (z'_y)^2} = \sqrt{2}$$

$\therefore \Sigma$ 的法向量为

$$\vec{n} = \{f'_x, f'_y, -1\} / \sqrt{2},$$

$$= \{\cos\alpha, \cos\beta, \cos\gamma\}$$



$$[D_{xy}: 1 \leq x^2 + y^2 \leq 4]$$

$$I = \iint_{\Sigma} y dy dz - x dz dx + z^2 dx dy$$

$$= \iint_{\Sigma} [y \cos \alpha - x \cos \beta + z^2 \cos \gamma] dS$$

$$= \iint_{\Sigma} - \frac{z^2}{\sqrt{1 + (z_x)^2 + (z_y)^2}} dS$$

$$= - \iint_{D_{xy}} (x^2 + y^2) dx dy \quad [D_{xy} : 1 \leq x^2 + y^2 \leq 4]$$

$$= - \int_0^{2\pi} d\theta \int_1^2 r^2 \cdot r dr = - \frac{15}{2} \pi.$$

课堂练习

一、填空题

1. $\Sigma : x^2 + (y-1)^2 = 1 (0 \leq z \leq 3)$

$$\iint_{\Sigma} (x^2 + y^2 - 2y + 5) dS = \underline{30\pi}$$

2. $\Sigma : (x-a)^2 + (y-b)^2 + (z-c)^2 = 1$ 的外侧

$$\text{则} \oiint_{\Sigma} x dy dz + y dz dx + z dx dy = \underline{4\pi}$$

3. $\oiint_{\Sigma(\text{外侧})} P dy dz + Q dz dx + R dx dy = \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dv$

成立的条件是 P, Q, R 有连续的一阶偏导数

4. Σ 为球面 $x^2 + y^2 + z^2 = a^2$, 则 $\oiint_{\Sigma} x dS = \underline{\quad 0 \quad}$

$\oiint_{\Sigma(\text{外侧})} x dy dz = \underline{\frac{4}{3} \pi a^3}$

二、 Σ 为 $z = \sqrt{x^2 + y^2}$ 被柱面 $x^2 + y^2 = 2x$ 所截得的部分, 其质量分布面密度为 $\rho(x, y, z) = zx$, 求此面对 z 轴的转动惯量 I_z .

解 $\Sigma : z = \sqrt{x^2 + y^2} \quad D_{xy} : x^2 + y^2 \leq 2x$

$$\begin{aligned} dS &= \sqrt{1 + \left(\frac{x}{\sqrt{x^2 + y^2}}\right)^2 + \left(\frac{y}{\sqrt{x^2 + y^2}}\right)^2} dx dy \\ &= \sqrt{2} dx dy \end{aligned}$$

$$I_z = \iint_{\Sigma} zx(x^2 + y^2) dS = \iint_{D_{xy}} x(x^2 + y^2)^{\frac{3}{2}} \sqrt{2} dx dy$$

$$= \sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} r \cos\theta r^3 dr$$

$$= \sqrt{2} \cdot \frac{2^6}{6} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^7\theta d\theta$$

$$= \sqrt{2} \cdot \frac{2^6}{3} \int_0^{\frac{\pi}{2}} \cos^7\theta d\theta$$

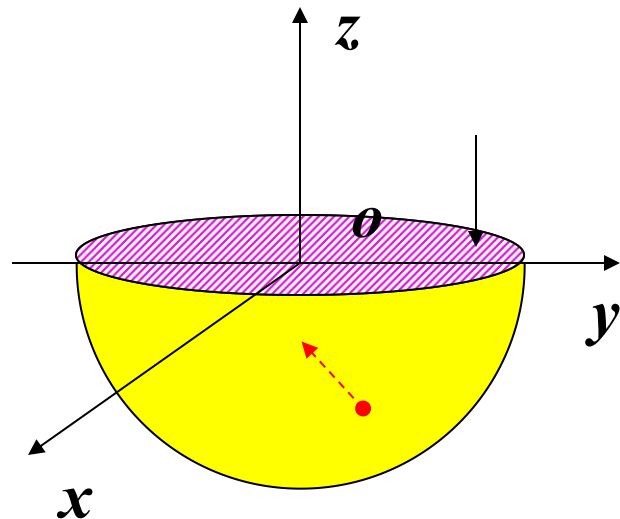
$$= \frac{\sqrt{2}}{3} 2^6 \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} = \frac{2^{10} \sqrt{2}}{105}$$

三、计算：
$$\iint_{\Sigma} \frac{axdydz + (z+a)^2 dx dy}{(x^2 + y^2 + z^2)^{\frac{1}{2}}} \quad (a > 0 \text{ 为常数})$$

其中 Σ 为下半球面 $z = -\sqrt{a^2 - x^2 - y^2}$ 的上侧，

解

$$\begin{aligned} & \iint_{\Sigma} \frac{axdydz + (z+a)^2 dx dy}{(x^2 + y^2 + z^2)^{\frac{1}{2}}} \\ &= \frac{1}{a} \iint_{\Sigma} axdydz + (z+a)^2 dx dy \\ &= \frac{1}{a} \left[\oiint_{\Sigma + \Sigma_1} - \iint_{\Sigma_1} axdydz + (z+a)^2 dx dy \right] \end{aligned}$$



补充 $\Sigma_1: z = 0$ (下侧),
 $(x, y) \in D: x^2 + y^2 \leq a^2$

$$= -\frac{1}{a} \iiint_{\Omega} [a + 2(z+a)] dv - \frac{1}{a} \iint_D (0+a)^2 dx dy = -\frac{\pi}{2} a^3$$

四.求 $\vec{v} = \vec{i} + z\vec{j} + \frac{e^z}{\sqrt{x^2 + y^2}}\vec{k}$ 穿过曲面 Σ 指定侧的流量,

其中 Σ 为 $z = \sqrt{x^2 + y^2}$ 介于 $z = 1, z = 2$ 之间的部分, 其法向量与 oz 轴的正向夹角为钝角

解

$$\Phi = \iint_{\Sigma} \vec{v} \cdot \vec{n} ds = \iint_{\Sigma} dydz + z dz dx + \frac{e^z}{\sqrt{x^2 + y^2}} dx dy$$

$$\iint_{\Sigma} dydz = \iint_{\Sigma_{\text{前}}} dydz + \iint_{\Sigma_{\text{后}}} dydz = 0,$$

$$\iint_{\Sigma} z dx dz = \iint_{\Sigma_{\text{左}}} z dx dz + \iint_{\Sigma_{\text{右}}} z dx dz = 0$$

$$\iint_{\Sigma} \frac{e^z}{\sqrt{x^2 + y^2}} dx dy = - \iint_{D_{xy}} \frac{e^{\sqrt{x^2 + y^2}}}{\sqrt{x^2 + y^2}} dx dy$$

$$= - \int_0^{2\pi} d\theta \int_1^2 \frac{e^r}{r} r dr = -2\pi(e^2 - e) \quad [D_{xy}: 1 \leq x^2 + y^2 \leq 4]$$

